

Guideline for System Development Documentation

For Event-Driven Programming (JavaScript)

1. Project Overview

- **System Title:**
- **Version:**
- **Prepared By:**
- **Date:**
- **Description:** A short summary of what the system does, its purpose, and target users.

2. Objectives

Add discussion here minimum of 5 sentences.

- Define the system's primary goals.
- State how event-driven programming is applied (e.g., event loops, listeners, async calls, UI interactivity).

3. Scope and Delimitation

Add discussion here minimum of 10 sentences.

3.1 Scope

Add discussion here minimum of 5 sentences.

- Features covered (e.g., user login, real-time event handling, notifications).
- Target users (e.g., students, staff, administrators).
- Supported platforms (web, mobile, desktop).
- Technologies/frameworks included.

3.2 Delimitation

Add discussion here minimum of 5 sentences.

- Features not covered (e.g., system does not support offline mode).
- Constraints (e.g., limited to Chrome/Edge browsers, requires internet connection).
- Limitations (e.g., event logs stored for 30 days only, handles up to 500 concurrent users).

4. System Architecture

Add discussion here minimum of 10 sentences.

4.1 Architectural Diagram

Add discussion here minimum of 10 sentences.

(Insert diagram of client-server interaction, event cycle, or component relationships)

4.2 Technologies Used

Add discussion here minimum of 10 sentences.

- **Programming Language:** JavaScript (Node.js / Frontend / Both)
- **Frameworks / Libraries:** Express.js, React/Vue.js, Socket.IO, etc.
- **Database:** MySQL, MongoDB, or Firebase
- **Tools:** Git, NPM, Webpack, etc.

5. Event-Driven Design

Add discussion here minimum of 10 sentences.

5.1 Event Flow Diagram

(Illustrate event lifecycle: event → listener → handler → response)

6. Database Design

Add discussion here minimum of 5 sentences.

6.1 ERD (Entity Relationship Diagram)

(Include diagram here)

Add discussion here minimum of 5 sentences.

7. Methods

Add discussion here minimum of 5 sentences.

7.1 Development Methodology

Add discussion here minimum of 10 sentences.

- Agile / Scrum / Waterfall (state methodology used).
- Iterative prototyping for UI and event handling.
- Continuous integration for testing event flows.

7.2 Programming Methods

Add discussion here minimum of 5 sentences.

- Use of **event listeners** for user/system interactions.

- **Callbacks/Promises/Async-Await** for async events.
- **Custom Events** for modular communication.

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